MARTINO FERRERO

Systems Engineer

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About me

I am a Systems Engineer passionate about solving complex software challenges, with a strong focus on backend development for applications powered by Artificial Intelligence and Machine Learning. My main interests lie in Large Language Models (LLMs) and Computer Vision, where I aim to design and implement scalable, efficient, and innovative solutions that bring real value across different domains.

Skilled In

Programming Languages	Python, JavaScript, TypeScript.
Backend Development	Flask, FastAPI, Node.js, Electron, REST, GraphQL, microservices.
Databases	PostgreSQL, MongoDB, ChromaDB, Redis.
ORM	Prisma, SQLAlchemy, Mongoose.
ML / AI	PyTorch, Tensorflow, Keras, scikit-learn, Pandas, Numpy, Matplotlib, OpenCV, NLTK, spaCy, LangChain, LangGraph, RAG, fine-tuning, CNNs, Transformers, OpenAI API, Hugging Face.
Cloud / Deployment	Docker, Kubernetes, CI/CD with Test Coverage, Virtual Machines (VirtualBox, VMware), AWS.
Tools / Practices / Others / Operating Systems	Git, GitHub, GitLab, Unit Testing (unittest, Pytest), Mocking (unittest.mock), Scrapping (BeautifulSoup), Bash Scripting, Development Methodologies (Agile/Scrum, Waterfall), UML, SOLID, Jira, Confluence, Kanban, Design and Architecture Patterns, Windows, Linux, MacOS.

Work Experience

PI Data Strategy & Consulting **Remote (2026–Present)**

AI Backend Engineer

Development of enterprise AI solutions and scalable backend platforms, combining software engineering with modern LLM-based architectures. Responsible for designing backend services, AI workflows, cloud deployments, and contributing to the architecture of production-ready AI systems.

- Designed and developed scalable backend services and REST APIs using Python and FastAPI, following modern software architecture and asynchronous programming practices.
- Built AI applications leveraging LLMs, agentic workflows, RAG pipelines, and prompt orchestration with LangChain and LangGraph.
- Collaborated on cloud-native deployments, infrastructure automation, and CI/CD pipelines while contributing to the design of scalable AI solutions within Microsoft Azure environments.
- Implemented observability, tracing, monitoring, and production best practices to improve the reliability, performance, and maintainability of AI systems.

Technologies: Python, FastAPI, LangChain, LangGraph, LLMs, Agentic AI, RAG, MCP, SDD, PostgreSQL, ChromaDB, Redis, Docker, Microsoft Azure, IaC, CI/CD, REST APIs, Observability.

INTIA - UNICEN **Tandil, Bs As (2026–Present)**

PhD Researcher in Medical Imaging and Artificial Intelligence

Pursuing a PhD focused on the application of artificial intelligence and medical imaging techniques for the early and accurate diagnosis of diseases associated with bone degradation, particularly osteoporosis.

- Conducting research on computer vision, deep learning, and medical image analysis methods to develop novel diagnostic approaches and improve the assessment of bone health using DXA and related imaging modalities.
- Collaborating with physicians, researchers, and multidisciplinary teams to design, validate, and translate AI-driven solutions into clinically relevant tools that support medical decision-making and patient care.
- Contributing to scientific research through the development, evaluation, and dissemination of innovative methodologies in the fields of medical imaging and artificial intelligence.

Technologies: Python, PyTorch, Computer Vision, Deep Learning, Medical Imaging.

Neurollow (Early-Stage Startup) **Tandil, Bs As (2025–Present)**

Founding Engineer

Development of a web-based application for the management, monitoring, and diagnostic evaluation of patients with Parkinson's disease, designed to support neurologists in clinical decision-making and streamline medical workflows. Neurollow is currently incubated through the IncuBAte startup incubation program.

- Designed and implemented backend services, RESTful APIs, and data models using TypeScript, Node.js, Express, Prisma, and PostgreSQL, supporting patient records, diagnoses, medications, and clinical assessments.
- Contributed to the integration of artificial intelligence and computer vision models for the automated evaluation of patient motor performance during UPDRS assessments, enabling objective analysis of movement patterns and supporting diagnostic processes.
- Collaborated with healthcare professionals and multidisciplinary teams to translate clinical requirements into scalable software solutions.

Technologies: TypeScript, Node.js, Express, Prisma ORM, PostgreSQL, Docker, REST APIs, Zod, Artificial Intelligence, Computer Vision.

Tradespark **Tandil, Bs As (2025)**

Algorithmic Trading Developer

Worked as part of a team developing custom algorithmic trading solutions for clients including banks and financial institutions. Focused on designing and implementing strategies to operate efficiently in capital markets, as well as creating internal automation tools to optimize workflows and ensure reliability of trading operations.

- Developed custom trading strategies leveraging Python and the ArQuants framework, tailored to client-specific requirements.
- Built internal tools to automate processes, improving efficiency and reducing manual effort.
- Participated in key stages of the development lifecycle, including design, implementation, testing, and deployment of trading solutions.
- Developed a QA automation bot to streamline testing on simulated markets, ensuring reliability and accuracy of trading strategies.
- Collaborated closely with cross-functional teams to translate client requirements into actionable, production-ready solutions.

Technologies: Python, ArQuants, Pandas, Automation Scripts, BeautifulSoup, Capital Markets APIs, CI/CD pipelines, Python-unittest.

INTIA - UNICEN **Tandil, Bs As (2024-2025)**

Computer Vision and Deep Learning Developer

Final-year project for my Bachelor's degree in Systems Engineering, focused on developing a comprehensive system for lumbar spine (L1–L4) segmentation on DXA anteroposterior images to enhance precision in medical evaluation. The work was documented in the article "Deep Learning-Based Segmentation of the Lumbar Spine in DXA for Improved Osteoporosis Assessment", accepted for presentation at the 21st International Symposium on Medical Information Processing and Analysis (SIPAIM) 2025 in Colombia and for publication in the IEEE Xplore Digital Library.

- Implemented traditional image processing techniques (OpenCV, NumPy, Matplotlib, scikit-image) alongside deep learning methods using U-Net-based neural networks developed in PyTorch, providing alternative segmentation strategies that improved multiple evaluation metrics compared to baseline approaches.
- Built a desktop application to integrate the entire workflow, leveraging Flask (backend), React (frontend), and Electron for deployment as a functional cross-platform system.
- Collaborated with medical experts to validate and refine the system, ensuring alignment with clinical requirements.

Technologies: Python, JavaScript, OpenCV, NumPy, Matplotlib, Feature Engineering, PyTorch, U-Net, Flask, React, Electron.

University Project / RAG-based Document Assistant **Tandil, Bs As (2024)**

AI & Backend Developer

Developed a Retrieval-Augmented Generation (RAG) application designed to allow users to query and interact with PDF documents in natural language. The system combined LangChain with the Google Gemini API, leveraging RAG to provide accurate and context-aware answers.

- Designed and implemented the RAG pipeline using LangChain, ensuring efficient document chunking, embedding, and retrieval.
- Integrated the solution with ChromaDB as the vector database to manage embeddings and optimize query performance.
- Connected the pipeline to the Gemini API, enabling high-quality generative responses grounded in retrieved context.

Technologies: Python, LangChain, ChromaDB, Gemini API, RAG.

Private Clinic **Capital Federal, Bs As (2023)**

Conversational AI Developer

Developed and deployed a WhatsApp-based chatbot to assist parents of pediatric patients, automating registration, appointment scheduling, and FAQs. Built with Rasa and Python, enhanced with NLTK, and integrated with clinic systems to manage medical staff availability and patient data.

- Integrated with Twilio WhatsApp API for secure communication.
- Deployed with Docker, Nginx, and PostgreSQL, ensuring scalability and reliability.
- Integrated the system with the clinic's platforms to provide dynamic access to doctors' schedules, professional availability, and patient data.
- Reduced administrative workload by automating repetitive tasks and improving patient experience.

Technologies: Python, Rasa, NLTK, Twilio WhatsApp API, PostgreSQL, Docker, Nginx.

University Project / Compiler Construction **Tandil, Bs As (2023)**

Java Developer

Developed a custom compiler as a university project, covering all main stages from lexical, syntactic, and semantic analysis to code generation in assembly.

- Implemented the syntactic analyzer with YACC, ensuring correct parsing and validation.
- Generated final assembly code, completing the end-to-end compiler workflow.

Technologies: Java, YACC, Assembly.

Education

University Diploma in Artificial Intelligence – *UNICEN, Argentina* (2025, GPA: 9.8/10)
Bachelor's Degree in Systems Engineering – *UNICEN, Argentina* (2020–2025, GPA: 8.6/10)
University Analyst-Programmer (Intermediate Degree) – *UNICEN, Argentina* (2020 – 2023)

Certificates

Fundamentals of Deep Learning - NVIDIA
Building Transformer-Based Natural Language Processing Applications - NVIDIA

Languages

Spanish: Native
English: Fluent